Roshan Ayyalasomayajula

Assistant Professor

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Work Experience

Fall 2023 – **Department of Computer Science and Engineering, University at Buffalo**, *Assistant Professor*.
Tenure track

April – July 2021 **Microsoft Research**, *Remote Intern*.
Topic: **Soil Parameter Estimation** | Mentors: Bodhi Priyantha

March – June 2020 **Nokia Bell Labs**, *Remote Intern*.
Topic: **Passive WiFi Sensing** | Mentors: Enrico Rantala, Swetha Muniraju

Education

2017-2023 **Doctor of Philosophy (Ph.D.)**, *University of California San Diego, Jacobs School of Engineering*, PI: Prof. Dinesh Bharadia..
Major: Signal and Image Processing.
- Thesis committee chaired by Dinesh Bharadia and with committee members: Deepak Vasisht, Peter Gerstoft, Nuno Vasconcelos, Xinyu Zhang, and Alex Snoeren.

2012-2016 **Bachelors of Technology (B.Tech.)**, *Indian Institute of Technology (IIT Roorkee)*, India.
Major: Electronics and Communication Engineering

Journal Publications

JASA Vol.150 Issue 2021 Yifan Wu, **Roshan Ayyalasomayajula**, MichaelBianco, Dinesh Bharadia, Peter Gerstoft. Sound source localization based on multi-task learning and image translation network. In The Journal of the Acoustical Society of America 150, 3374 (2021); doi: 10.1121/10.0007133

IMWUT Vol.5 Issue 3 2021 Minghui Zhao, Tyler Chang, Aditya Arun, **Roshan Ayyalasomayajula**, Chi Zhang, Dinesh Bharadia. ULoc: Low-Power, Scalable and cm-Accurate UWB-Tag Localization and Tracking for Indoor Applications. In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 5.3 (2021): 1-31.

Conference Publications

In Review Duong, Long, Charuvahan Adhivarahan, **Roshan Ayyalasomayajula**, and Karthik Dantu. TIPS: Thermal Image based Plastics Sorting. In review to Mobisys 2026.

Arxiv 2026 Chen, Wenhao, Wenyi Morty Zhang, Wei Sun, Dinesh Bharadia, and **Roshan Ayyalasomayajula**. SpyDir: Spy Device Localization Through Accurate Direction Finding. In submission. Arxiv link: `https://arxiv.org/abs/2602.00411`

Arxiv 2025 Roy, Kanishka*, Tahsin Fuad Hasan*, Chenfeng Wu, Eshwar Vangala, and **Roshan Ayyalasomayajula**. FedWiLoc: Federated Learning for Privacy-Preserving WiFi Indoor Localization. In submission. Arxiv link: `https://arxiv.org/abs/2512.18207`

Arxiv 2024 Wei Sun, Aditya Arun, Vaibhav Anand, **Roshan Ayyalasomayajula**, Dinesh Bharadia. DOLOS: Tricking the Wi-Fi APs with Incorrect User Locations. In submission. Arxiv link: `https://arxiv.org/abs/2407.16138`

MobiSys'24 Aditya Arun, William Hunter, **Roshan Ayyalasomayajula**, Dinesh Bharadia. WAIS: leveraging WiFi for resource-efficient SLAM. In Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services 2024 Jun 3 (pp. 561-574).

RA-L/ICRA 2022 Aditya Arun, **Roshan Ayyalasomayajula**, Chenfeng Wu, Joel Bisarra, Tyler Chang, Dinesh Bharadia. P^2SLAM: Two-Way Bearing-based WiFi SLAM for Indoor Navigation. In submission to The 2021 International Conference on Robotics and Automation (ICRA 2021). IEEE, Xi'an, China

ICASSP 2021 Yifan Wu, **Roshan Ayyalasomayajula**, Michael Bianco, Dinesh Bharadia, Peter Gerstoft. Blind Sound Source Localization based on Deep Learning. In proceedings of the 46th edition of The international Conference on Acoustics, Speech, & Signal Processing (ICASSP'21).IEEE, Toronto, Canada

NSDI 2020 **Roshan Ayyalasomayajula**, Srivatsan Rajagopalan, Aditya Arun, Shreya Ganesaraman, Aravind Seetharman, Chenfeng Wu and Dinesh Bharadia. LocAP: Accurate Localization of Existing WiFi Infrastructure . In proceedings of 17th USENIX Symposium on Networked Systems Design and Implementation(NSDI'20).USENIX, Santa Clara, CA, USA

Mobicom 2020 **Roshan Ayyalasomayajula**, Aditya Arun, Chenfeng Wu, Sanatan Sharma, Abhishek Sethi, Deepak Vasisht, and Dinesh Bharadia. Deep learning based wireless localization for indoor navigation. In Proceedings of the 26th Annual International Conference on Mobile Computing and Networking.ACM, London, UK.

CONEXT 2018 **Roshan Ayyalasomayajula**, Deepak Vasisht, and Dinesh Bharadia. BLoc: CSI-based Accurate Localization for BLE Tags . In Proceedings of International Conference on emerging Networking EXperiments and Technologies (CoNEXT'18).ACM, New York, NY, USA.

CVIP 2017 **Roshan Ayyalasomayajula** and Pankajakshan, V., 2017. Differentiating Photographic and PRCG Images Using Tampering Localization Features. In Proceedings of International Conference on Computer Vision and Image Processing (pp. 429-438). Springer, Singapore.

Workshop Publications

AAAI ML4Wireless 2026 Hasan, Tahsin Fuad, Ayush Roy, **Roshan Ayyalasomayajula**, and Vishnu Suresh Lokhande. Attending to Routers Aids Indoor Wireless Localization." In AAAI 2026 Workshop on Machine Learning for Wireless Communication and Networks (ML4Wireless).

WiNTECH 2024 Pratyaksh Mundra*, Zhengyu Huang*, Dharmi Khadela, Prachi Sinha, William Hunter, Aditya Arun, Dinesh Bharadia, and **Roshan Ayyalasomayajula**. WiSenseHub: Architecture to deploy a building-scale WiFi-sensing system. To appear in the 18th ACM Workshop on Wireless Network Testbeds, Experimental evaluation & Characterization 2024, Washington, D.C., USA.

HotMobile 2023 **Ayyalasomayajula, Roshan**, Aditya Arun, Wei Sun, and Dinesh Bharadia. "Users are closer than they appear: Protecting user location from WiFi APs." In Proceedings of the 24th International Workshop on Mobile Computing Systems and Applications, pp. 124-130. 2023.

Patents

US2025 0328144A1 Arun, Aditya, **Roshan Ayyalasomayajula**, William Hunter, and Dinesh Bharadia. Device localization and navigation using rf sensing. U.S. Patent Application 18/872,190, filed October 23, 2025.

US2024 0118367A1 Zhao M, **Ayyalasomayajula SR**, Zhang C, Bharadia D, Arun A, Chang T, inventors. Ultra-wideband localization. United States patent application US 18/257,568. 2024 Apr 11.

US11140651B2 **Ayyalasomayajula SR**, Bharadia D, Vasisht D, Katabi D, inventors. Location determination of wireless communications devices. United States patent application US 16/731,738. 2020 Jul 2.

US12282111B2 **Ayyalasomayajula SR**, Bharadia D, Ganesaraman S, Jain I, Rajagopalan S, Seetharaman A, Sharma S, Arun A, Wu C, inventors. Wireless device localization. United States patent application US 17/604,380. 2022 Jun 23.

Thesis

Spring 2023 **Ayyalasomayajula, Sai Roshan**. ML Based Wireless Sensing Systems for Robotics and IoT Applications. University of California, San Diego, 2023.

Grants & Funding

Dec 2025 - Feb 2026 SUNY AI Platform – "Federated and Differential-Privacy deployments for Indoor Wi-Fi sensing"

Teaching

Spring 2026 **Instructor**, *University at Buffalo, CSE 711-AYY – AI-Driven Wireless Sensing and Next-Generation Communication Systems*.

Spring 2024, 2025 **Instructor**, *University at Buffalo, CSE 4/510-AYY – Algorithms for Mobile and IoT Devices.*

Fall 2023, 2024, 2025 **Instructor**, *University at Buffalo, CSE 4/589 – Modern Networking Concepts*.

Winter 2020 **Teaching Assistant**, *UCSD, ECE 257B – Principles of Wireless Networks or Modern Wireless Communications*.
Helped the professor schedule the classes, design and grade assignments and examinations.

Professional Services

July 2024 – **CSExplore**.
To design and organize CSExplore Camp for K-12 and Robotics Day for community outreach.
Summer 2025, 2026 – Camp Director; Summer 2024 – Camp's Organizing Committee

Jan 10, 2024 **Invited Talk**.
IEEE COMSOC UP Section & IEEE Roorkee Sub-Section: Talk titled "ML-Based Wireless Sensing for Robotics and IoT Aplications"

2022 – **Program Committee (PC)**.
SenSys'25 ENSsys'25 Mobicom'25 NSDI'24 Mobicom'23 Artefact Evaluation: *TPC* **(mmNets) with Mobicom'23**, **(ICA3PP) 2023 and 2024**, **ACM SenSys 2022**: *Shadow PC*

2022 **Presented and organized Kaggle compeition and WILD datasets**.
1st International workshop on Wireless AI perception, CVPR 2022

2019 – **Technical Reviewer**.
ICC'25 SAC Aerial Communications Track; IEEE Transactions on Robotics, Transactions on Antennas and Propagation, Transactions on mobile Computing, Transactions on Networking, Robotics and Automation Letters, International Conference on Robotics and Automation,Transactions on Instrumentation and Measurement, Journal of Indoor and Seamleass Positioning and Navigation. ACM Journal on Interactive, Mobile, Wearable and Ubiquitous Technologies